

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Equation of a Straight Line

A line is controlled by gradient and one point.

EQUATION OF A STRAIGHT LINE · P1 · 08

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 GRAD: Find The Gradient

TRIGGER *two points, parallel line, perpendicular line*

02 POINT: Build The Line

TRIGGER *gradient and point are known*

03 PARA: Parallel And Perpendicular

TRIGGER *parallel or perpendicular*

04 MEET: Intersection Of Lines

TRIGGER *where two lines meet*

BANK EVIDENCE Local pages: 26 problem, 57 answer, 18 notes. Website primary entries: 17.

01 GRAD: Find The Gradient

TRIGGER *two points, parallel line, perpendicular line*

BECOMES Gradient is change in y over change in x .

FIRST LINE TO WRITE

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

SIMPLEST STRATEGY

- 1 Get two points.
- 2 Rise is change in y .
- 3 Across is change in x .
- 4 Divide rise by across.

WORKED MODEL

$$m = \frac{7 - 1}{4 - 2} = 3.$$

HARD VARIANTS

- 1 Parallel gradients are equal; perpendicular gradients are negative reciprocals.
- 2 If coordinates contain letters, calculate the gradient symbolically first.
- 3 Vertical lines have undefined gradient and need $x = \text{constant}$.

— BOTTOM LINE

Gradient is a ratio, not just a difference.

02 POINT: Build The Line

TRIGGER *gradient and point are known*

BECOMES Use point-gradient form.

FIRST LINE TO WRITE

$$y - y_1 = m(x - x_1)$$

SIMPLEST STRATEGY

- 1 Pick the known point.
- 2 Obtain the gradient.
- 3 Insert into point-gradient form.
- 4 Neaten into $y = mx + c$ if needed.
- 5 Test with the point.

WORKED MODEL

$$m = 2, (3, 5) : y - 5 = 2(x - 3) \Rightarrow y = 2x - 1.$$

HARD VARIANTS

- 1 If the point is an intercept, convert it to coordinates before using the formula.
- 2 For parameter lines, substitute the given point to find the parameter.
- 3 Return to the requested form: $y = mx + c$, $ax + by + c = 0$, or coordinate form.

— BOTTOM LINE

One point plus one gradient fixes the line.

03 PARA: Parallel And Perpendicular

TRIGGER *parallel or perpendicular*

BECOMES Parallel gradients match; perpendicular gradients multiply to -1 .

FIRST LINE TO WRITE

$$m_{\parallel} = m, \quad m_{\perp} = -\frac{1}{m}$$

SIMPLEST STRATEGY

- 1 Pick the original gradient.
- 2 Adapt it for parallel/perpendicular.
- 3 Replace into line equation.
- 4 Answer in required form.

WORKED MODEL

| If $m = 4$, a perpendicular line has gradient $-\frac{1}{4}$.

HARD VARIANTS

- 1 For perpendicular lines, invert and change sign before building the new line.
- 2 If the original line is not in gradient form, rearrange it first.
- 3 Use the given point after finding the new gradient.

— BOTTOM LINE

Perpendicular means negative reciprocal.

04 MEET: Intersection Of Lines

TRIGGER *where two lines meet*

BECOMES Set the two equations equal.

FIRST LINE TO WRITE

$$m_1x + c_1 = m_2x + c_2$$

SIMPLEST STRATEGY

- 1 Make y-values equal.
- 2 Evaluate x .
- 3 Evaluate y .
- 4 Turn into coordinate form.

WORKED MODEL

$$2x + 1 = 5 - x \Rightarrow 3x = 4, x = \frac{4}{3}.$$

HARD VARIANTS

- 1 If the lines are parallel, there may be no meeting point.
- 2 With parameters, solve the intersection condition to find the unknown first.
- 3 Give the answer as a coordinate and verify both equations.

— BOTTOM LINE

The meeting point is an ordered pair.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
two points, parallel line, perpendicular line	$m = \frac{y_2 - y_1}{x_2 - x_1}$
gradient and point are known	$y - y_1 = m(x - x_1)$
parallel or perpendicular	$m_{\parallel} = m, \quad m_{\perp} = -\frac{1}{m}$
where two lines meet	$m_1x + c_1 = m_2x + c_2$